

Clinical Scoring Systems for Adults With Acute Appendicitis

Natasha Pyzocha, DO

CLINICAL QUESTION

Which clinical scoring system is most accurate for risk stratification of acute appendicitis in adults?

EVIDENCE SUMMARY

Acute appendicitis is the most common cause of abdominal pain and requires urgent surgical intervention, with a lifetime risk between 7% and 12% in the general population.¹ Diagnosis remains a challenge because only about 40% of patients present with classic symptoms of pain migration to the right lower quadrant, nausea, and vomiting.¹ Delayed diagnosis increases the risk of perforation and peritonitis, whereas overdiagnosis leads to unnecessary appendectomy, which occurs in approximately 10% to 15% of cases.²

The Alvarado score was introduced in 1986 and is the most widely studied tool for identifying acute appendicitis.³ It incorporates six clinical features and two laboratory variables³ (Table 1^{1,2}). Cutoffs differ by study, but generally, a score of less than 4 makes appendicitis unlikely, 4 to 6 indicates possible acute appendicitis and imaging is recommended, and scores of 7 or greater suggest the need for early surgical consultation.⁴

The RIPASA score was developed to improve diagnostic accuracy for acute appendicitis by incorporating demographic factors and additional physical examination findings not included in the Alvarado score.⁵ It includes 14 parameters, and a score of 7.5 or greater is considered

TABLE 1

Alvarado and RIPASA Scores for Acute Appendicitis

Category	Alvarado (points)	RIPASA (points)
Demographics	—	Age: ≤ 40 years (1); > 40 (0.5) Sex: female (0.5); male (1)
Laboratory findings	Left shift (1) Leukocytosis (2)	Leukocytosis (1) Negative urinalysis (1)
Signs	Fever (1) Rebound pain (1) Right lower quadrant tenderness (2)	Fever (1) Guarding (2) Rebound pain (1) Right lower quadrant tenderness (1) Rovsing sign (2)
Symptoms	Anorexia (1) Migration of pain (1) Nausea or vomiting (1)	Anorexia (1) Duration of symptoms ≤ 48 hours (1) > 48 hours (0.5) Migration of pain (0.5) Nausea or vomiting (1) Right lower quadrant pain (0.5)
Cutoff score	≥ 7	≥ 7.5

Note: Leukocytosis is defined as a white blood cell count > 10,000/μL ($10 \times 10^9/L$), and a left shift is defined as > 75% neutrophils.

Information from references 1 and 2.

NATASHA PYZOCHA, DO, Oscar Medical Group, New York, New York; *AFP* Contributing Editor

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Address correspondence to Natasha Pyzocha, DO, at natashapyzocha@gmail.com.

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positive and should prompt surgical consultation¹ (Table 1^{1,2}). The RIPASA score was most extensively validated in Middle Eastern and Asian populations, although subsequent studies suggest reasonable performance in more diverse settings.² The original derivation study included a variable for identifying foreign nationals, reflecting the heterogeneous population in Singapore; this variable is often omitted in external validations and calculators used in other regions.⁵

Several additional scoring systems, including the Appendicitis Inflammatory Response, Adult Appendicitis Score, Fenyö-Lindberg score, Lintula, and Tzanakis, have been proposed. However, most of these scores have limited external validation

TABLE 2

Accuracy of Alvarado and RIPASA Scores for Diagnosis of Acute Appendicitis in Patients With Acute Abdominal Pain

Score	Sensitivity (95% CI)	Specificity (95% CI)	LR+	LR–	Pretest probability of appendicitis	Probability of appendicitis	
RIPASA	93% (78%-98%)	81% (62%-92%)	4.90	0.086		≥ 7.5 points	< 7.5 points
					10%	35%	1.0%
					20%	55%	2.1%
					40%	77%	5.4%
Alvarado	77% (68%-83%)	72% (61%-80%)	2.75	0.32		≥ 7 points	< 7 points
					10%	23%	3.3%
					20%	41%	7.4%
					40%	65%	18%

Note: Scores are dichotomized using commonly studied cutoffs; results reflect pooled estimates across heterogeneous validation studies.

LR+ = positive likelihood ratio; LR– = negative likelihood ratio.

Information from reference 2.

or inconsistent performance across populations, and none have demonstrated clear superiority to RIPASA or Alvarado among adults in routine practice.²

Table 2 summarizes the probability of appendicitis based on the RIPASA and Alvarado scores.² A 2022 systematic review and meta-analysis of 33 studies (n = 5,384) found that the RIPASA score had a significantly higher sensitivity than the Alvarado score (95% vs 72%) and slightly lower specificity (71% vs 77%).¹ A subsequent 2025 meta-analysis of 40 studies (n = 9,370) confirmed greater overall diagnostic accuracy for the RIPASA score and reported a higher sensitivity (93% vs 77%), specificity (81% vs 72%), and area under the curve (0.91 vs 0.79; $P < 0.002$).² Because of its higher sensitivity, the RIPASA score may be more useful for ruling out appendicitis when negative. However, substantial heterogeneity across studies and the greater information burden required for RIPASA (14 vs 8 items) may limit its clinical applicability.

APPLYING THE EVIDENCE

A 24-year-old man has had abdominal pain that migrates to the right lower quadrant for the past 18 hours. He has nausea but no anorexia. Physical examination reveals right lower quadrant

tenderness and guarding without rebound pain. Rovsing sign is negative. He is afebrile. Laboratory testing shows leukocytosis without a left shift and negative urinalysis.

Using the Alvarado score, the patient receives 6 points, placing him in the intermediate-risk category, and confirmatory imaging is recommended. Using the RIPASA score, the patient receives 10 points, which should prompt a surgical consultation.

REFERENCES

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